

Assignment 2

1.

Item	Mass, Kg	Price, \$
1	6	20
2	4	50
3	5	10
4	3	30

Maximum Mass = 13 kg

The above problem is the 0/1 knapsack problem. You have to maximize the price so that the maximum mass does not exceed 13 kg. Now you are asked to use a Genetic Algorithm for this problem.

- Encode the problem and create an initial population of 4 chromosomes
- Explain the fitness function appropriate for this problem. Use that to find the two fittest chromosomes
- Using the selected chromosomes, perform a single crossover to find two offspring
- Perform mutation and check the fitness again.

2.

Object	Reward	Weight
A	20	1
B	5	2
C	10	3
D	40	8
E	15	7
F	25	4
G	4	5
H	7	2

The above problem is a 0/1 Knapsack problem. You have to carry the different objects in your bag in a way such that the reward is maximized without exceeding the weight limit. You can carry an object exactly once, but you always have to carry the object labeled “H”. Assuming you are asked to use a Genetic Algorithm for this problem, answer the following questions

1. Encode the problem and create an initial population of 4 different chromosomes
2. Explain what would be an appropriate fitness function for this problem. Use the fitness function and perform natural selection to choose the 2 fittest chromosomes.
3. Using the selected chromosomes, perform a single-point crossover to get 2 offspring.
4. Perform mutation and check the fitness of the final offspring. Explain your work.

3.

Assume a set of **10 numbers** represented by Y1,Y2,Y3,Y4,Y5,Y6,Y7,Y8,Y9,Y10 where each number can range from **1 to 100**.

Your task is to **find a combination** of numbers such that the **absolute difference** between the sum of **prime numbers** and the sum of **composite numbers** in the set is **100**.

Let:

- **P** = the sum of all **prime** numbers in the set
 - **C** = the sum of all **composite** numbers in the set
- Then:
- $|P-C|=100$

You have to solve this problem using a **Genetic Algorithm**.

4.

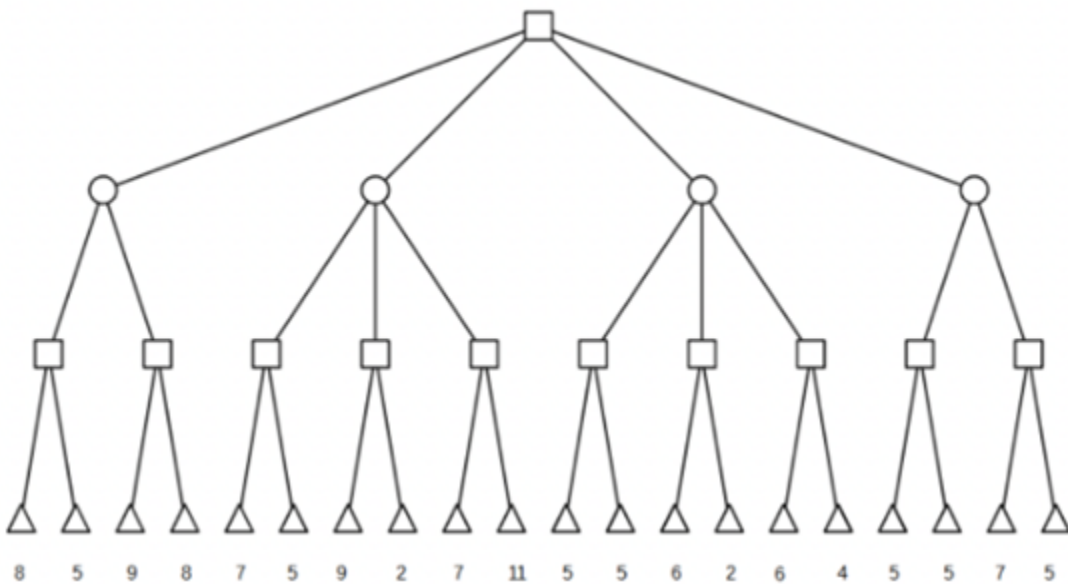
You are trying to break a digital lock that uses a **6-digit PIN code**. The lock has the following **secret constraints**:

1. **The sum of all 6 digits must be exactly 30.**
2. **No two adjacent digits can be equal.**
3. **The number must contain exactly two even digits.**
4. **The PIN must start with an odd number.**

Your task is to **use a Genetic Algorithm** to evolve a chromosome (PIN code) that satisfies **all four constraints**.

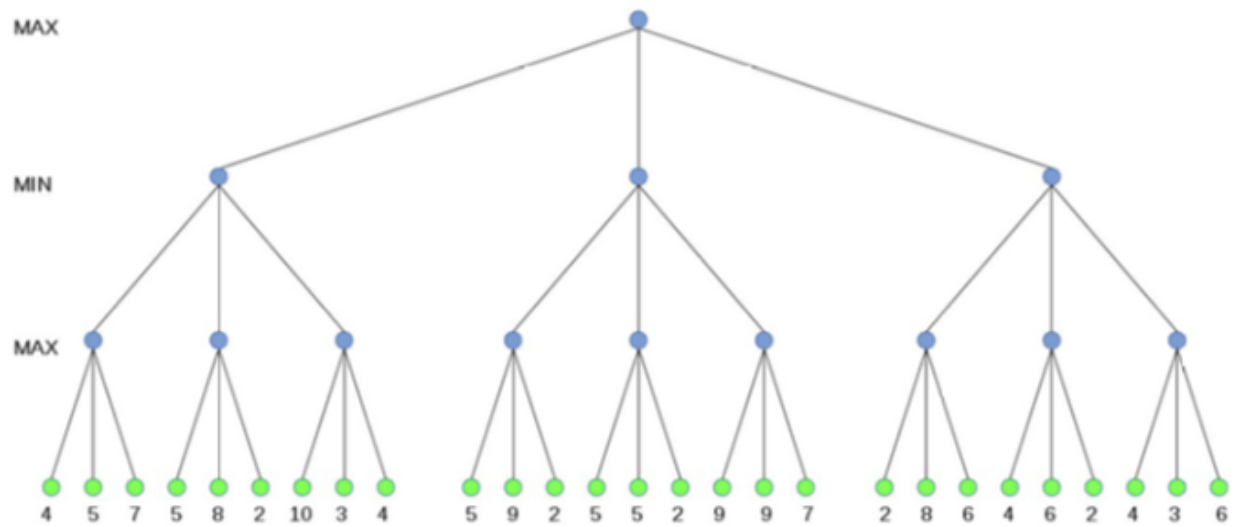
5. Showcase a graph of five nodes where the heuristic is admissible but inconsistent.
6. State and explain the issues with the hill-climbing algorithm. Incorporate the idea of the 8-Queens problem to relate to the issues of hill climbing

7.



What will be the alpha- and beta-values of each node in this tree if alpha-beta pruning is run on this tree? Also, illustrate the crossed-out branches that would be pruned by alpha-beta pruning.

8.



What will be the alpha- and beta-values of each node in this tree if alpha-beta pruning is run on this tree? Also, illustrate the crossed-out branches that would be pruned by alpha-beta pruning.